Confidence Interval Estimator

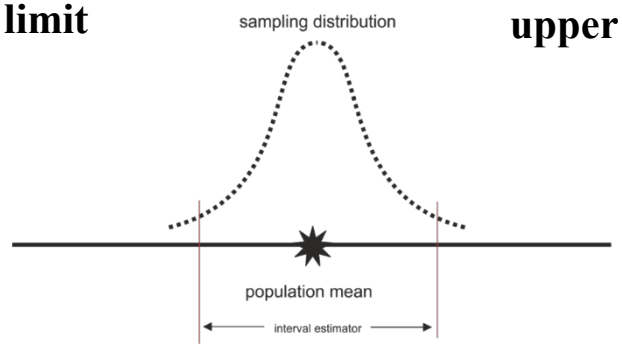
- **Even the most efficient point estimator is unlikely to estimate population parameter exactly.**
- **It is true that the estimation accuracy increases with the large samples, but it doesn't guarantee that a point estimation is equal to population parameter.**
- **Interval estimator: Start with selecting a confidence level**

Confidence level: This value defines your sureness level that indicates the value you found in estimation is lying on the interval.

How confident are you that the population parameter is lying on btw 5 and 15?
*Are you **99%** sure? **90%** sure?*

$$\mu_{Xmin} \leq \mu_X \leq \mu_{Xmax}$$

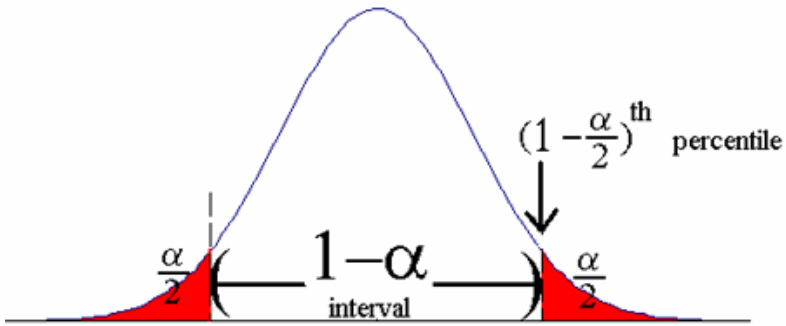
lower confidence limit **upper confidence limit**



Development of Interval

- Interval estimator for μ_X is an interval in the form of $\mu_{Xmin} \leq \mu_X \leq \mu_{Xmax}$
- Consider we can determine the values of μ_{Xmin} and μ_{Xmax} such that the following probability is true:

$$P(\mu_{Xmin} \leq \mu_X \leq \mu_{Xmax}) = 1 - \alpha$$



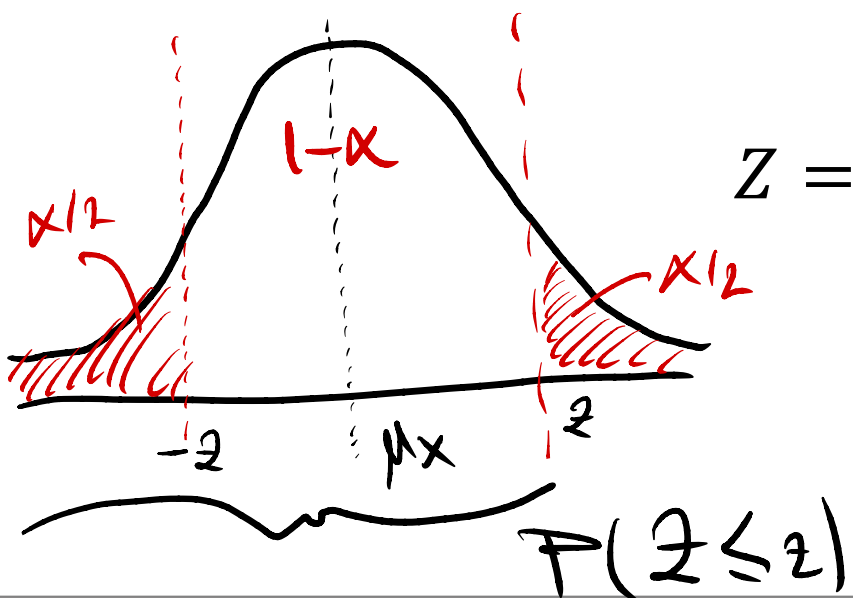
- $(1-\alpha)$ is called the confidence coefficient/level

- When α=0.05, we have a 95% confidence level and when α=0.01, we have a 99% confidence level
- The wider confidence level is the more confident we can be that the invertal contains the unknown parameter.

CI for the mean of a normal distribution: population variance known

- Suppose that $X_1, X_2, \ldots. X_n$ are random samples with unknown population mean (μ_X) and known variance (σ_X).
- We know a fact that (from CLT), the sample mean ($\bar{X}$) is normal dist. with μ_X and σ_X

$$P(-z \leq Z \leq z) \Rightarrow P\left(-z \leq \frac{\bar{X} - \mu_X}{\sigma_X/\sqrt{n}} \leq z\right)$$



$$Z = \frac{\bar{X} - \mu_X}{\frac{\sigma_X}{\sqrt{n}}}$$

$$P\left(-z\sigma_X/\sqrt{n} \leq \bar{X} - \mu_X \leq z\sigma_X/\sqrt{n}\right)$$

$$P\left(\bar{X} - z\sigma_X/\sqrt{n} \leq \mu_X \leq \bar{X} + z\sigma_X/\sqrt{n}\right)$$

$$= 1 - \alpha$$

$UL = \bar{X} + z\sigma_X/\sqrt{n}$ $\quad LL = \bar{X} - z\sigma_X/\sqrt{n}$

Standard Normal CDF

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Example

The life in hours of a 75 watt light bulb is known to be normal distribution with population std dev. 20 hours. A random sample of 30 bulbs has a mean life of sample mean 1014 hours.

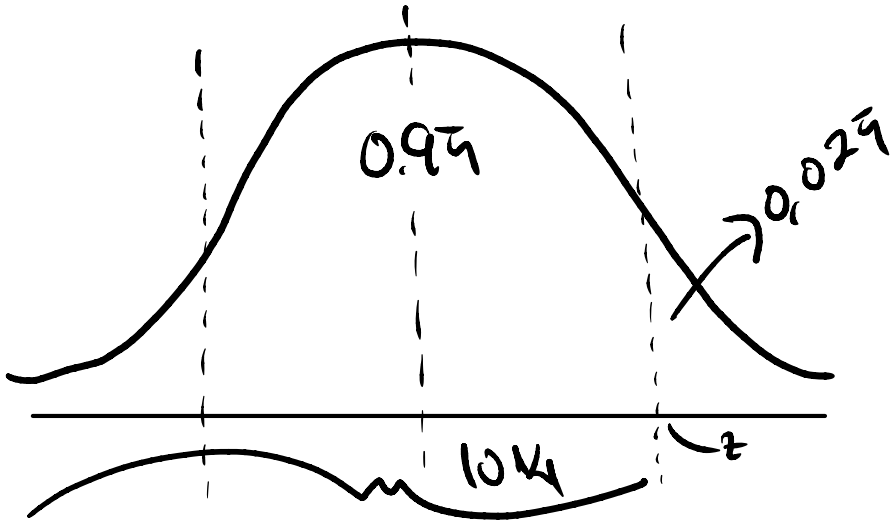
a) Construct a 95% confidence interval on the mean life.
b) Construct a 95% lower confidence bound on the mean life.
c) Suppose that we wanted the error in estimating the mean life from the two-sided confidence interval to be six hours at 95% confidence. What sample size should be used?

a) 95% Two-sided CI on the true mean life of a 75-watt light bulb
For $\alpha = 0.05$, $z_{\alpha/2} = z_{0.025} = 1.96$, and $\bar{x} = 1014$, $\sigma = 20$, $n=30$

$$\bar{x} - z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right) \le \mu \le \bar{x} + z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$1014 - 1.96\left(\frac{20}{\sqrt{30}}\right) \le \mu \le 1014 + 1.96\left(\frac{20}{\sqrt{30}}\right)$$

$$1006.84 \le \mu \le 1021.16$$



$P(Z \le z) = 1 - 0.025 = 0.975$

$z = 1.96$

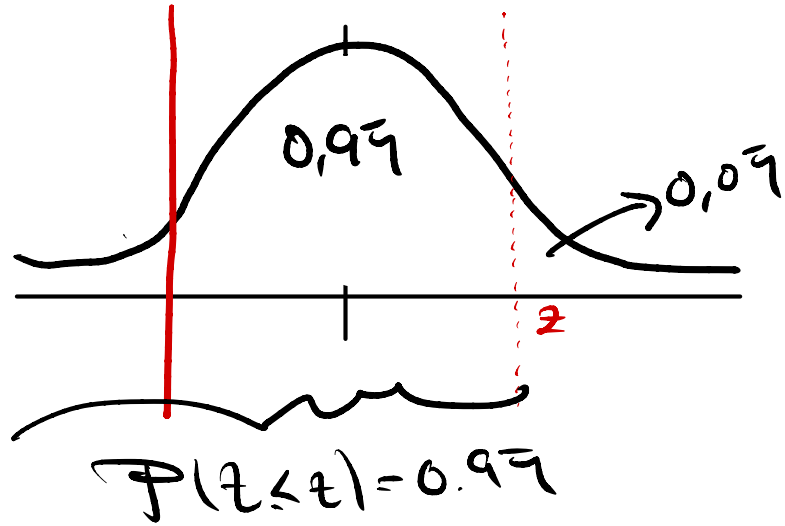
$UL = \bar{x} + z \cdot \sigma_x / \sqrt{n} = 1014 + 1.96 \cdot \frac{20}{\sqrt{30}} = 1021.16$ | $LL = 1014 - 1.96 \times \frac{20}{\sqrt{n}} = 1006.84$

b) 95% one-sided CI on the true mean piston ring diameter
For $\alpha = 0.05$, $z_\alpha = z_{0.05} = 1.65$ and $\bar{x} = 1014$, $\sigma = 20$, n=30

$$\bar{x} - z_{0.05}\frac{\sigma}{\sqrt{n}} \le \mu$$

$$1014 - 1.65\left(\frac{20}{\sqrt{30}}\right) \le \mu$$

$$1007.98 \le \mu$$



$z = 1.65 \longrightarrow LL = 1014 - z.\sigma_x/\sqrt{n} \Rightarrow 1007.98$

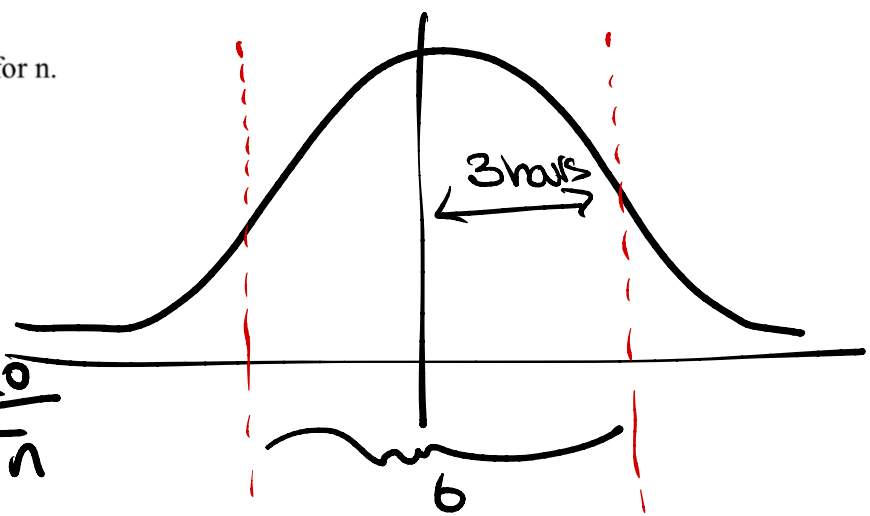
Set the width to 6 hours with $\sigma = 20$, $z_{0.025} = 1.96$ solve for n.

1/2 width $= (1.96)(20)/\sqrt{n} = 3$

$39.2 = 3\sqrt{n}$

$n = \left(\frac{39.2}{3}\right)^2 = 170.74$

Therefore, n = 171



$3 = z.\sigma_x/\sqrt{n} \Rightarrow 3 = 1.96.\frac{20}{\sqrt{n}}$

$z = 1.96$

Development of Interval

You'll probably be finding confidence intervals using the Normal. But in reality, most confidence intervals are found using the t-distribution (especially if you are working with small samples and sample variance). Development of an interval can be examined in to 2 different categories.

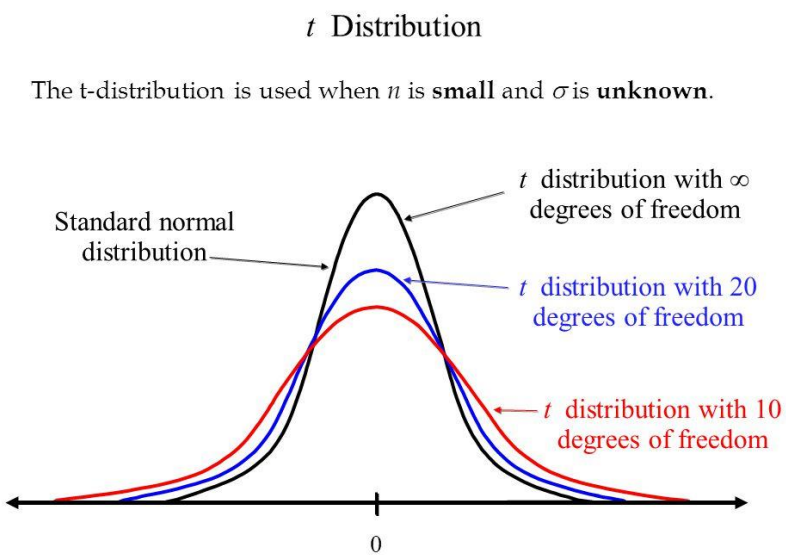
If you aren't sure which technique you should use.

population variance	sample variance
Z dist.	t dist.

t-distribution

What do we do if we don't know σ? If the population of interest is normally distributed, we can use the Student's t-distribution in place of the standard normal distribution.

t-distribution is any member of a family of continuous probability distributions. The t-distribution looks like the normal distribution, except that it is more spread out. It is still symmetrical about the mean; mean=median=mode; goes from -∞ to +∞.



$$z = \frac{\bar{X} - \mu_X}{\frac{\sigma_X}{\sqrt{n}}}$$

$$t = \frac{\bar{X} - \mu_X}{\frac{\bar{\sigma}_X}{\sqrt{n}}}$$

Confidence Level for t distribution

t-distribution (especially if you are working with population standard deviation is unknown))

Process: How to find the confidence interval (min and max value)

General method to derive the confidence interval for t-distribution

Step 1: DoF: Subtract 1 from your sample size.

Step 2: Confidence Coef: Subtract the confidence level from 1, then divide by two.

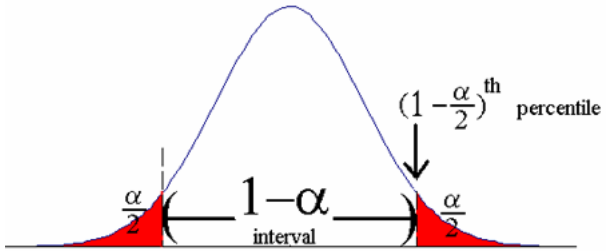
Step 3: Look up your answers to step 1 and 2 in the t-distribution table.

Step 4: Divide your sample standard deviation by the square root of your sample size.

Step 5: Margin of Error: Multiply step 3 by step 4.

Step 6: For the lower end of the range, subtract step 5 from the sample mean.

Step 7: For the upper end of the range, add step 5 to the sample mean.



$$\text{Confidence interval}$$

$$\bar{X} \pm t\frac{\overline{\sigma_X}}{\sqrt{n}}$$

Example

A group of 10 foot surgery patients had a **sample mean weight of 240** pounds. The **sample standard deviation was 25** pounds. Find a two sided confidence interval for a sample for the true mean weight of all foot surgery patients. Find a 95% Confidence Interval.

Alpha	*0.250*	*0.200*	*0.150*	*0.100*	*0.050*	*0.025*	*0.010*	*0.005*	*0.0005*
df									
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.656	636.578
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.600
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850

$DOF = 10 - 1 = 9$

$\alpha/2 = (1 - 0.95)/2 = 0.025$

$t(DOF, \alpha/2) = 2.262$

$UL = \bar{x} + t\,\sigma_{\bar{x}}/\sqrt{n} \Rightarrow 240 + 2.262 \times 25/\sqrt{10} = 257.883$

$LL \Rightarrow 240 - 2.262 \times 25/\sqrt{10} = 222.11$

Example

$$\mu_{\bar{x}} = \frac{\sum x_i}{10} = 71 \qquad \sigma_{\bar{x}} = \sqrt{\frac{\sum (x_i - \mu_{\bar{x}})^2}{9}} = 18.172$$

Construct a two-sided 98% Confidence Interval based on the following sample data from normal distribution: 45, 55, 67, 45, 68, 79, 98, 87, 84, 82.

Sample mean= 71,
Sample standard dev.=18.172

Step 1: DoF= 9
Step 2: COEF [redacted] = 0.01
Step 3: 2,821
Step 4: 18.172/sqrt(10)=5.75
Step 5: 2.821 x 5.75=16.22
Step 6: 71-16.22=54.77
Step 7: 71+16.22=87.22

Alpha	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.0005
df									
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.656	636.578
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19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850

$$UL = 71 + 2.821 \cdot \frac{18.172}{\sqrt{10}} = 87.22$$

$$LL = 71 - 2.821 \cdot \frac{18.172}{\sqrt{10}} = 54.77$$

Example

A confidence interval estimate is desired for the gain in a circuit on a semi conductor device. Assume that gain is normal distribution with **population standard deviation 25**

a. Find a 95% CI for population mean when n=10 and the sample mean 1000
b. How large must n be if the MoE of the two-sided 95% CI is to be 30?
c. How large must n be if the MoE of the two-sided 99% CI is to be 30?

a) 1000 +1.96(25/ sqrt(10)= 1015.5
1000 -1.96(25/ sqrt(10)= 984.5

b)

c)